

PROBLEM STATEMENT #09 | Campus & Academic Operations

Faculty Research Grant & Publication Tracker

1. Problem Context & Overview

The research deanery requires an automated portal to track sponsored research grants, faculty publication metrics (indexing, citations), co-author approval workflows, and fund burn-up analytics.

Target Stakeholders / Actors: Faculty Researcher, Research Dean

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall track grant fund allocations, allow faculty to log equipment procurement expenses, and compute remaining budgetary balance.

Sample Acceptance Criteria: Pass: Balance updates accurately and blocks expenses exceeding grant cap; Fail: Overdraft allowed without admin override.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: The system shall maintain an immutable, tamper-evident audit ledger for all financial approvals and journal status modifications.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- Complete Requirements Table:
 - >> Exactly 5 Functional Requirements (FR-001 to FR-005)
 - >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.
- UML Use-Case Diagram:
 - >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.
- Use-Case Flow Specification:
 - >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
 - >> One Alternate Flow.